ZCAS University
School of Computing Technology and Applied Sciences
Department of Computer Science

CIT 2641 INTRODUCTION TO DATABASES

MID TERM TEST

TIME ALLOWED: ONE AND HALF HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. There are three case studies in this paper. You are required to attempt any TWO cases from three provided.
2. Candidates must not turn this page until the invigilator tells them to do so.
3. Each Case study Carries 25 Marks
4. You are required to answer each Question on a new page

CASE STUDY 1: University Student Information Management System

A rapidly growing university is facing challenges in managing student records, course registrations, and academic performance tracking. The institution has decided to develop a centralized database system using MySQL to improve efficiency and data integrity. As a database developer, you are tasked with designing and implementing this system from scratch. You are required to identify key entities such as students, lecturers, courses, departments, and enrollments, and support operations like student registration, course enrollment, grade recording, and transcript generation. The system must ensure consistency, reduce redundancy, and allow efficient querying of academic data.

(a) Identify and clearly describe at least **five entities** and their attributes

(b) Develop an **Entity-Relationship (ER) diagram** showing:

a) Entities, attributes, and relationships

b) Cardinalities and participation constraints

(c) Transform your ER diagram into **relational tables**, clearly indicating: **i)** Primary keys

ii) Foreign keys

- Normalization (up to **3NF**)

(e) Write SQL statements to Create the database, tables and apply constraints (PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE)

CASE STUDY 2: Smart Hospital Management System

a) Consider the following information and draw the ER Schema diagram using **Min** and **Max** notation:

Alternative (min, max) notation for relationship structural constraints:

- Specified on each participation of an entity type E in a relationship type R
- Specifies that each entity e in E participates in at least *min* and at most *max* relationship instances in R
- Default(no constraint): min =0, max=n (signifying no limit)
- Must have min ≤max, min ≥0, max ≥1
- Derived from the knowledge of mini -world constraints
- Examples:
  - A department has exactly one manager and an employee can manage at most one department.
    - Specify (0,1) for participation of EMPLOYEE in MANAGES
    - Specify (1,1) for participation of DEPARTMENT in MANAGES
  - An employee can work for exactly one department but a department can have any number of employees.
    - Specify (1,1) for participation of EMPLOYEE in WORKS_FOR
    - Specify (0,n) for participation of DEPARTMENT in WORKS_FOR

b) Consider the following table containing some of the currently Available Automated Database Design Tools information. Fill the table with the appropriate functionality for every tool.

COMPANY	TOOL		FUNCTIONALITY
Embarcadero Technologies	**ER Studio**		
	DB Artisan		
Oracle	**Developer 2000/Designer 2000**		
Popkin Software	**System Architect 2001**		
Platinum (Computer Associates)	**Enterprise Modeling Suite: Erwin, BPWin, Paradigm Plus**		
Persistence Inc.	**Pwertier**		
Rational (IBM)	**Rational Rose**		
Resolution Ltd.	**Xcase**		
Sybase	**Enterprise Application Suite**		
Visio	**Visio Enterprise**		

CASE STUDY 3: Company Database

The company is organized into Departments. Each department has a name, number, and an employee who manages the department. We keep track of the start date if the department manager. A department may have several locations. Each department controls a number of Projects. Each project has a name, number, and is located at a single location. We store each Employees social security number, address, salary, sex, and birth date. Each employee works for one department but may work on several projects. We keep track of the number of hours per week that an employee currently works on each project. We also keep track of the direct supervisor of each employee. Each employee may have a number of Dependents. For each dependent, we keep their name company, set, birth date, and relationship to the employee.

Task: You are required to Conceptualize this scenario.